PCSE211 GROUP PROJECT — PART 1

Problem and Feasibility: Egg Production Tracking System

Brenda's Poultry Farm, Muyenga, Kampala

1. Problem statement

Brenda's Poultry Farm in Muyenga keeps chickens for egg production. The farm has about 70 laying hens and collects around 50–60 eggs a day. One worker records production in a large counter book, and Brenda, the owner, checks the entries and confirms the totals.

The main problem is that production totals are added by hand. This can lead to calculation mistakes and make it hard to know the correct number of eggs produced each day or week. It can also make it difficult to compare production over time. The worker who records the eggs and Brenda, who checks the figures, are both affected.

The group proposes a small smartphone-based Egg Production Tracking System. A worker will enter the daily number of eggs collected and record damaged eggs separately. The system will calculate daily and weekly totals automatically. It will focus only on egg production records, not sales, general inventory or flock health. This project is about analysis and design; no coding is required.

2. Chosen development model: Iterative model

The group will use the iterative model. This means planning a simple version, showing it to the lecturer, getting feedback, and improving the design in stages.

- The lecturer will check each part, so feedback can be used before moving to the next stage.
- The system is small, so its screens and functions can be reviewed and changed without redesigning everything.

3. Alternatives compared

Option	Advantages	Disadvantages
Keep using the counter book	No new equipment or training is needed. Staff already know it.	Totals still need to be added by hand, so mistakes can continue.
Use a spreadsheet on a phone	Can calculate totals automatically and may be low-cost.	A spreadsheet may be awkward to use on a small phone screen.
Small smartphone tracking system	Designed for the farm's egg records; calculates totals and keeps records together.	Needs setup, basic training and a way to back up records.

4. Feasibility study (TELOS)

Area	Assessment	Reason
Technical	Possible	A simple phone-based system can use entry forms and automatic totals. The farm needs a suitable smartphone.
Economic	Likely affordable	Free or low-cost tools may be used. Setup, training and backup costs should be checked.
Legal	Likely feasible	The system will mainly store egg production figures, and there is no obvious legal barrier to recording these digitally. If it stores workers' names or other personal details, the farm should protect that information, limit access to authorized users, and follow Uganda's Data Protection and Privacy Act, 2019.
Operational	Possible	A worker already records production and Brenda checks it. Short training and simple screens should help.
Schedule	Feasible	The group can prepare the design in stages and show each part to the lecturer for feedback.

5. Cost-benefit analysis and payback period3

The amounts below are estimates for planning, not prices confirmed by the farm. They assume the farm already has a suitable smartphone and that a low-cost tool can be used.

Estimated item	Cost (UGX)
Initial setup and configuration	150,000
Basic user training	50,000

Total initial cost	200,000
Backup/support per month	5,000

Possible benefits are less time spent adding figures, fewer calculation errors and easier access to production totals. For an example, if the system saves one hour a day for 26 days a month, and that time is valued at UGX 3,000 per hour, the time saved is worth UGX 78,000 per month. After UGX 5,000 monthly support, the estimated net benefit is UGX 73,000 per month.

Payback period = initial cost ÷ monthly net benefit = UGX 200,000 ÷ UGX 73,000 = about 2.7 months. This is only an estimate based on assumptions. The group should check whether the time saved and costs are reasonable.

6. Three main risks

Risk	How to reduce it
Wrong figures entered	Keep the form simple and let Brenda check the totals regularly.
Worker finds the system difficult	Use clear labels and give the worker a short practice session.
Records are lost or phone is unavailable	Back up records regularly and keep the counter book as a temporary backup during changeover.

7. Recommendation

The group recommends planning a small smartphone-based Egg Production Tracking System. Compared with the counter book, it can calculate daily and weekly totals automatically and help reduce manual addition errors. The iterative model will let the group improve the design after feedback from the lecturer and farm users. Before use, the farm should confirm the estimated costs, smartphone availability and backup method.